

NanoVDB `ReadAccessor` Benchmark — OLD vs NEW, CPU vs GPU

Benchmarks comparing the `NANOVDB_USE_OLD_ACCESSOR` caching behaviour (ON vs OFF) across access patterns, on CPU (single- and multi-threaded) and GPU. Results are collected from two machines: a **laptop** and a **desktop**.

What is being measured

The `ReadAccessor` caches the tree nodes visited on the previous access so a spatially nearby access can skip the traversal from the root.

- **OLD** (`NANOVDB_USE_OLD_ACCESSOR` defined): on a leaf-cache miss the level-1 and level-2 cache checks are compiled away (`if constexpr + else`), so the accessor falls straight to a full **root traversal**.
- **NEW** (`NANOVDB_NO_OLD_ACCESSOR` defined): all applicable cache levels are always checked, so a leaf-cache miss can still hit the **level-1 / level-2** cache instead of restarting at the root.

Methodology

What the numbers measure

Every reported figure is the **wall-clock time of a single random-access read** into the grid, i.e. one `accessor.getValue(ijk)` call, expressed in **nanoseconds per lookup** (`ns/lookup`). It is a pure *read-path* micro-benchmark: no values are written, no math is done on the result beyond a running sum (kept `volatile` / written to an output array so the compiler cannot elide the work).

- **CPU-1T** — one thread walks the coordinate list serially with a single accessor. This is a **latency** number: the cost of one dependent lookup.

- **CPU-MT** — the same total work split across all hardware threads (`nanovdb::util::forEach` → `tbb::parallel_for`), each grain using its own accessor. This is a **throughput** number: total time ÷ total lookups with all cores busy.
- **GPU** — the coordinate list is uploaded to the device; each thread processes a 32-coord chunk with its own accessor. Timed with CUDA events (kernel time only, excluding the one-time grid upload). Also a **throughput** number.

The only thing that differs between the OLD and NEW columns is the accessor's cache-fallback logic (a compile-time switch); the data, coordinates, and harness are identical.

The test data

A **fog-volume sphere** built with

```
nanovdb::tools::createFogVolumeSphere<float>
(radius=500, voxelSize=1.0, halfWidth=3.0) . Its
```

measured properties:

Property	Value
Value type	<code>float</code>
Index bounding box	<code>[-500, -500, -500] ... [500, 500, 500]</code> (1001 ³ voxels)
Dense voxel count of bbox	1,003,003,001 (~1.00 B)
Active voxels	523,592,077 (~523.6 M)
Fill ratio (active / dense bbox)	52.2 %
Upper internal nodes (level 2, 32 ³)	8
Lower internal nodes (level 1, 16 ³)	272
Leaf nodes (level 0, 8 ³)	82,608
Grid size in memory	188,484,640 bytes (~180 MB)

So it is a **fully populated solid ball** — the interior is all active voxels (the 52 % fill ratio is just the volume of a sphere inside its cube, $\pi/6 \approx 0.524$). It is *sparse* in the VDB sense (only the ball

is stored, not the empty corners), but the region the benchmark actually samples is **100 % active**: all access patterns are confined to a cube well inside radius 500, so every lookup lands on a real leaf and traverses the tree for real — we are **not** measuring empty-space shortcuts, we are measuring genuine node traversal and cache reuse.

How much work, how it is averaged

Point patterns (Sequential / LeafJump / NodeJump / Random)	1,048,576 lookups each
Stencil	262,144 centres × 27 neighbours = 7,077,888 lookups
Sampled region	cube of half-extent 256 (points) / $[0, 64)^3$ dense cube (stencil), both fully inside the ball
Repetition	each configuration run 7 times; the median is reported (robust to OS jitter / turbo ramp)
Warm-up	GPU does one untimed warm-up launch before the 7 timed trials

Hardware / build

Two machines were benchmarked; results for each are in separate sections below.

	Laptop	Desktop
CPU	32 HW threads; TBB <code>parallel_for</code> , 4096-coord grains	AMD Ryzen 9 9950X — 16 cores / 32 threads; same TBB settings
GPU	NVIDIA RTX 5000 Ada Generation Laptop GPU (SM 8.9); 32-coord chunks/thread, 128 threads/block	NVIDIA RTX PRO 6000 Blackwell Max-Q Workstation Edition (SM 12.0); same CUDA settings

	Laptop	Desktop
Build	Release (-O3 -DNDEBUG), C++17 / CUDA 17	same
OLD vs NEW	selected at compile time per target (-DNANOVDDB_USE_OLD_ACCESSOR vs -DNANOVDDB_NO_OLD_ACCESSOR)	same

Access patterns

Pattern	Stride / shape	Cache behaviour
Sequential	1	leaf (level-0) cache always hot
LeafJump	8 (= leaf dim)	leaf cache cold; level-1 warm (NEW only)
NodeJump	128 (= lower-node dim)	leaf + level-1 cold; level-2 warm (NEW only)
Random	uniform in cube	all caches cold — both fall to root
Stencil	3×3×3 = 27 neighbours per centre, dense sweep	mostly same-leaf hits; boundary neighbours spill to level-1 (NEW rescues)

Results — Desktop (AMD Ryzen 9 9950X / RTX PRO 6000 Blackwell, SM 12.0)

Combined — ns per lookup (one `getValue` call), lower = faster

The stencil row is the per-lookup cost inside a 27-neighbour sweep.

Pattern	CPU-1T OLD	CPU-1T NEW	CPU-MT OLD	CPU-MT NEW	GPU OLD	GPU NEW
Sequential (stride 1)	2.45	1.44	0.26	0.18	0.035	0.020
LeafJump (stride 8)	2.06	1.41	0.21	0.16	0.035	0.020

Pattern	CPU-1T OLD	CPU-1T NEW	CPU-MT OLD	CPU-MT NEW	GPU OLD	GPU NEW
NodeJump (stride 128)	1.85	1.64	0.22	0.21	0.028	0.024
Random (uniform)	7.19	9.93	0.47	0.57	0.039	0.047
Stencil (3×3×3, 27-pt)	2.143	1.371	0.170	0.102	0.0587	0.0275

Legend: CPU-1T = single thread · CPU-MT = 32 threads (TBB)
· GPU = RTX PRO 6000 Blackwell.

OLD → NEW speedup (>1 = the fix is faster)

Pattern	CPU-1T	CPU-MT	GPU
Sequential	1.70×	1.44×	1.75×
LeafJump	1.46×	1.31×	1.75×
NodeJump	1.13×	1.05×	1.17×
Random	0.72×	0.82×	0.83×
Stencil	1.56×	1.67×	2.14×

Detailed tables — Desktop

CPU, single-threaded (latency) — ns per lookup

Pattern	OLD	NEW	OLD → NEW
Sequential	2.45	1.44	1.70×
LeafJump	2.06	1.41	1.46×
NodeJump	1.85	1.64	1.13×
Random	7.19	9.93	0.72×
Stencil	2.143	1.371	1.56×

CPU, 32 threads (throughput) — ns per lookup

Pattern	OLD	NEW	OLD → NEW	MT speedup (NEW)
Sequential	0.26	0.18	1.44×	8.0×
LeafJump	0.21	0.16	1.31×	8.8×
NodeJump	0.22	0.21	1.05×	7.8×
Random	0.47	0.57	0.82×	17.4×
Stencil	0.170	0.102	1.67×	13.4×

GPU (throughput) — ns per lookup

Pattern	OLD	NEW	OLD → NEW
Sequential	0.035	0.020	1.75×
LeafJump	0.035	0.020	1.75×
NodeJump	0.028	0.024	1.17×
Random	0.039	0.047	0.83×
Stencil	0.0587	0.0275	2.14×

Stencil (3×3×3) — reported per whole 27-neighbour stencil

Platform / Mode	OLD ns/stencil	NEW ns/stencil	OLD → NEW
CPU 1 thread	57.86	37.01	1.56×
CPU 32 threads	4.58	2.74	1.67×
GPU	1.5862	0.7413	2.14×

Fair platform comparison — NEW accessor, full hardware (ns per lookup)

Workload	CPU-1T	CPU-MT	GPU	GPU vs CPU-MT
Sequential	1.44	0.18	0.020	9.0×
Random	9.93	0.57	0.047	12.1×
Stencil	1.371	0.102	0.0275	3.7×

Results — Laptop (RTX 5000 Ada Generation Laptop GPU, SM 8.9)

Combined — ns per lookup (one `getValue` call), lower = faster

The stencil row is the per-lookup cost inside a 27-neighbour sweep.

Pattern	CPU-1T OLD	CPU-1T NEW	CPU-32T OLD	CPU-32T NEW	GPU OLD	GPU NEW
Sequential (stride 1)	3.57	1.96	0.56	0.46	0.058	0.058
LeafJump (stride 8)	3.82	2.21	0.55	0.49	0.057	0.057
NodeJump (stride 128)	3.94	3.05	0.58	0.53	0.054	0.055
Random (uniform)	11.47	15.32	0.82	1.04	0.088	0.095
Stencil (3×3×3, 27-pt)	4.61	2.13	0.322	0.191	0.086	0.040

Legend: CPU-1T = single thread · CPU-32T = 32 threads (TBB)
· GPU = RTX 5000 Ada.

OLD → NEW speedup (>1 = the fix is faster)

Pattern	CPU-1T	CPU-32T	GPU
Sequential	1.82×	1.22×	1.00×
LeafJump	1.73×	1.12×	1.00×
NodeJump	1.29×	1.09×	0.98×
Random	0.76×	0.79×	0.93×
Stencil	2.17×	1.68×	2.17×

Detailed tables — Laptop

CPU, single-threaded (latency) — ns per lookup

Pattern	OLD	NEW	OLD → NEW
Sequential	3.57	1.96	1.82×
LeafJump	3.82	2.21	1.73×
NodeJump	3.94	3.05	1.29×
Random	11.47	15.32	0.76×
Stencil	4.61	2.13	2.17×

CPU, 32 threads (throughput) — ns per lookup

Pattern	OLD	NEW	OLD → NEW	MT speedup (NEW)
Sequential	0.56	0.46	1.22×	4.3×
LeafJump	0.55	0.49	1.12×	4.5×
NodeJump	0.58	0.53	1.09×	5.7×
Random	0.82	1.04	0.79×	14.7×
Stencil	0.322	0.191	1.68×	11.1×

GPU (throughput) — ns per lookup

Pattern	OLD	NEW	OLD → NEW
Sequential	0.058	0.058	1.00×
LeafJump	0.057	0.057	1.00×
NodeJump	0.054	0.055	0.98×
Random	0.088	0.095	0.93×
Stencil	0.086	0.040	2.17×

Stencil (3×3×3) — reported per whole 27-neighbour stencil

Platform / Mode	OLD ns/stencil	NEW ns/stencil	OLD → NEW
CPU 1 thread	124.48	57.49	2.17×
CPU 32 threads	8.70	5.17	1.68×
GPU	2.32	1.07	2.17×

Fair platform comparison — NEW accessor, full hardware
(ns per lookup)

Workload	CPU-1T	CPU-32T	GPU	GPU vs CPU-32T
Sequential	1.96	0.46	0.058	7.9×
Random	15.32	1.04	0.095	10.9×
Stencil	2.13	0.191	0.040	4.8×

Cross-machine comparison (Laptop vs Desktop)

1. The stencil win is universal and consistent

The NEW accessor's stencil speedup is **~2× on GPU and 1.6–2.2× on CPU across both machines**. This is the most reproducible finding — it does not depend on CPU generation or GPU architecture.

2. GPU point-pattern speedup is architecture-dependent — the most interesting finding

Pattern	Laptop Ada (SM 8.9)	Desktop Blackwell (SM 12.0)
Sequential	1.00×	1.75×
LeafJump	1.00×	1.75×
NodeJump	0.98×	1.17×
Random	0.93×	0.83×

On Ada, the extra cache-level checks in NEW are a complete wash for point patterns — the GPU compiler cannot exploit the conditional hierarchy check under warp divergence. On **Blackwell (SM 12.0) the same code extracts 1.75× for coherent point patterns**, suggesting the newer SM handles predication / branch overhead much more efficiently.

3. Desktop CPU is faster in absolute terms but shows smaller relative gains

The Ryzen 9950X is 1.4–1.6× faster per-core in absolute ns. But the NEW accessor's per-lookup speedup is slightly *smaller* on the desktop (e.g. stencil 1T: **2.17× laptop vs 1.56× desktop**). A faster baseline leaves less relative room for the cache-level improvement to show.

4. Random always regresses — hardware-independent in character

The regression is intrinsic: all caches miss anyway and the extra checks are pure overhead. It is roughly **0.72–0.83× on both machines**, confirming this is a property of the access pattern, not the hardware.

5. The qualitative pattern is fully reproducible across machines

Same ranking, same sign on every row across both machines. The two independent data sets confirm each other's conclusions — the results are not artifacts of a single system.

Bottom line: the fix is solidly beneficial and portable. Stencil is the universal ~2× win. The point-pattern GPU benefit is an architectural bonus that appears on Blackwell (SM 12.0) but not on Ada (SM 8.9), and is the only result that varies qualitatively between the two machines.

Per-machine takeaways

- **Stencil access is the biggest and most universal win**, consistently **~2× on GPU** and **1.6–2.2× on CPU** across both machines. Each centre does 27 correlated lookups; boundary-crossing neighbours that the OLD accessor sends to the root are caught by the NEW accessor's level-1 cache, and the saving compounds across the 27.
- **Coherent point access benefits on CPU** (1.1–1.8× depending on machine and thread count). The desktop (Blackwell GPU, SM 12.0) also shows **~1.75× GPU speedup** for sequential / LeafJump patterns — a difference from the laptop (Ada, SM 8.9) where the GPU was neutral (1.00×) for those patterns. The Blackwell

micro-architecture appears to expose more of the cache-level benefit on the GPU path.

- **Only pathological fully-random access regresses slightly** ($\sim 0.7\text{--}0.8\times$ on CPU, $\sim 0.83\text{--}0.93\times$ on GPU), because every cache level misses anyway and the extra checks are pure overhead.
- **The multi-threaded CPU makes the GPU comparison fair.** Against a fully loaded 32-core CPU the GPU is $\sim 4\text{--}12\times$ faster (not the misleading $30\text{--}170\times$ a single thread would suggest). The gap is smallest for the stencil, where heavy cache reuse plays to the CPU's larger per-core caches.
- **CPU multi-thread scaling is memory-bound:** only $\sim 8\times$ on 32 cores for coherent patterns on the desktop (vs $\sim 4\text{--}5\times$ on the laptop), reflecting the desktop CPU's higher per-core bandwidth. Random / stencil scale better ($\sim 17\times / \sim 13\times$) because their higher per-access latency gives the cores more to overlap.

Building & running

```
# Configure with the benchmark (and CUDA, for the GPU
cmake -S . -B build -DNANOVDDB_BUILD_BENCHMARK=ON -DNA

# Build all four executables
cmake --build build --target \
  bench_accessor_old bench_accessor_new \
  bench_accessor_cuda_old bench_accessor_cuda_new -j

# Run (CPU prints single- and multi-threaded columns;
cd build/nanovdb/nanovdb/benchmark
./bench_accessor_old          # CPU, OLD accessor
./bench_accessor_new          # CPU, NEW accessor
./bench_accessor_cuda_old     # GPU, OLD accessor
./bench_accessor_cuda_new     # GPU, NEW accessor
```

The OLD vs NEW behaviour is selected at compile time per target (`-DNANOVDDB_USE_OLD_ACCESSOR` vs `-DNANOVDDB_NO_OLD_ACCESSOR`), so no reconfigure is needed to compare them.

Files

File	Purpose
BenchPatterns.h	Shared access-pattern generation (identical coords for CPU & GPU)
BenchAccessor.cc	CPU benchmark (single- and multi-threaded)
BenchAccessorCuda.cu	GPU (CUDA) benchmark
CMakeLists.txt	Builds the OLD/NEW × CPU/GPU targets